

RASPBERRY PI 5 - 40-PIN HEADER ALLOCATION & CONNECTIONS

PIN COLOUR KEY

- 3.3V Power (3V3)
- 5V Power (5V)
- GND (common ground bus)
- I2C Bus (SDA / SCL)
- Hall Sensor Inputs (GPIO)
- Available GPIO / PWM / SPI (reserved)
- EEPROM ID (do not use)

I2C ADDRESS MAP

Address	Device	Status	Notes
0x20	MCP23017 U1	INSTALLED	A0=A1=A2=open
0x21	MCP23017 U2	INSTALLED	A0 bridged
0x22	MCP23017 U3	INSTALLED	A1 bridged
0x23	MCP23017 U4	SPARE/DNF	A0+A1 bridged
0x40	(available)	-	PCA9685 removed from design
0x41	(available)	-	PCA9685 removed
0x42-4F	(available)	-	reserved for expansion
0x60	PCA9685 (alt)	ALT ADDRESS	ALLCALL=0x70

SENSOR GPIO ALLOCATION TABLE

GPIO	Pin	Sensor ID	Block	Direction
GPIO4	7	S0	Block A	ENTRY
GPIO17	11	S1	Block A	EXIT
GPIO18	12	S2	Block B	ENTRY
GPIO27	13	S3	Block B	EXIT
GPIO22	15	S4	Block C	ENTRY
GPIO23	16	S5	Block C	EXIT
GPIO24	18	S6	Block D	ENTRY
GPIO25	22	S7	Block D	EXIT
GPIO5	29	S8	spare	(DNF)
GPIO6	31	S9	spare	(DNF)
GPIO16	36	S10	spare	(DNF)
GPIO19	35	S11	spare	(DNF)

RPi 40-PIN HEADER (J8)

ODD	EVEN
PINFUNCTION	PINFUNCTION
39 GND	40 GPIO 21

ALLOCATION NOTES

I2C BUS (SDA/SCL) - Pins 3 & 5

Connects via TXB0104 LS1 level shifter

Drives U1/U2/U3 MCP23017 (0x20/21/22)

U4 spare (0x23) on same bus

I2C: U1/U2/U3 MCP23017 (0x20/21/22) . ADS1115 (0x48)

HALL EFFECT SENSOR INPUTS - GPIO 4,17,18,27,22,23,24,25

8 dedicated GPIO inputs, internal PUD_UP enabled

KY-024 DO pin -> GPIO, active LOW on magnet

VCC from 3V3 (Pin 1 or 17), GND common

Max cable run 10m at 3.3V with PUD_UP

PWM OUTPUT - GPIO 12 (PWM0 -> TRACTION)

Hardware PWM -> Traction PWM pin (output enable)

GPIO 13 (PWM1) spare hardware PWM - e.g. accessory lighting

SPI PINS - Reserved (not used)

GPIO 7-11 reserved in case SPI device added later

POWER BUDGET

Raspberry Pi (idle): ~500mA @ 5V

MCP23017 x 3: ~3mA each = 9mA

TXB0104 LS1: ~1mA

PCA9685 x 2: ~10mA each = 20mA

KY-024 sensors x 8: ~8mA total

Total 5V draw: ~800mA+ (Pi 5 idle ~620mA, PS1 2A OK)

PHYSICAL NOTES

Use 40-pin header riser or IDC ribbon to enclosure

Pi GPIO are 3.3V only - never apply 5V directly

All GND pins are equivalent - use multiple to reduce ground impedance on long cable runs

Pin 1 = 3V3 corner nearest USB-C power port (Pi 5)

SPARES POLICY

Spare Pi 5 stored in anti-static bag, SD/NVMe cloned annually. Pi 5 uses USB-C PD (27W/5A). Do NOT use a phone charger - use official 27W PSU or >=5A PD supply.

- 3V3 Power
- 5V Power
- GND
- I2C (SDA/SCL)
- Sensor GPIO
- Available GPIO
- DNF = Do Not Fit (spare)

PROJECT

G Scale Garden Railway Control

Raspberry Pi GPIO Allocation

SHEET

D01 / D08

RPi Header Detail

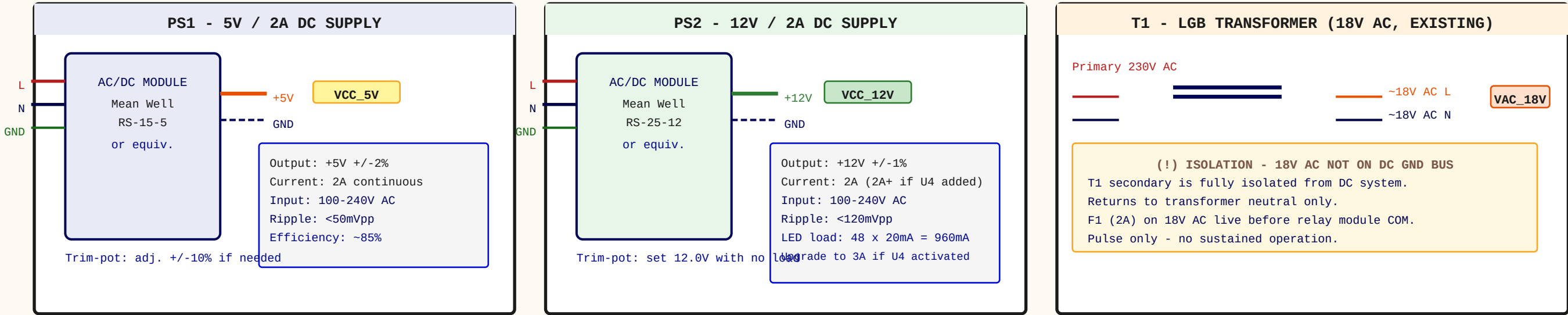
REV

C

DATE

2026-03

POWER SUPPLY STACK - PS1 (5V) . PS2 (12V) . T1 (18V AC) - DETAIL + SPARES



FUSE PROTECTION DETAIL

F-MAINS (in mains lead / plug fuse)

5A

 BS1362 13A plug fuse or DIN rail fuse holder 5x20mm
Protects entire system from mains fault

F2 - Spare Fuse Holder (DNF)

Spare holder fitted. Value TBD on expansion.
May be used for 12V DC branch fuse if load increases.

F1 - 18V AC Line Fuse

2A

 5x20mm glass fuse, 2A, in DIN rail fuse holder F1
On 18V AC live rail, before relay module COM terminal
Protects against wiring fault in turnout motor circuit

SPARE / BACKUP SUPPLIES (store in enclosure spare bay)

PS1-SPARE - 5V / 2A (identical to PS1)

5V/2A
SPARE
DNF

 Same part number as PS1.
Keep pre-wired with AC terminals so swap is <5 min in field.
Label: "PS1 SPARE - 5V 2A"

SPARE FUSES (store in labelled bag)

x 8 5x20mm 2A glass (F1 replacements)
x 4 5x20mm 5A glass (F-MAINS spares)
Store in small labelled zip-lock bag.
Bag location: inside enclosure, cable gland side.
Note: check fuse BEFORE suspecting PSU failure.

PS2-SPARE - 12V / 3A (uprated from PS2)

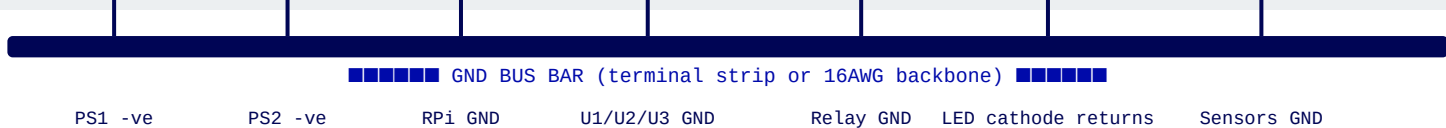
12V/3A
SPARE
DNF

 Uprated to 3A - serves as spare AND as upgrade path when U4 expander is activated (64 LEDs).
Label: "PS2 SPARE - 12V 3A"

DC BARREL JACKS - J1 (5V), J2 (12V)

5.5/2.1mm centre-positive barrel jacks.
Screw terminal breakout boards for solid connection. Torque to 0.4Nm max.
Polarity: centre pin = +V, sleeve = GND.
Spare jacks x2 kept in enclosure.

COMMON GROUND BUS - GND_MAIN



(!) T1 18V AC neutral does NOT connect to this bus. Isolated return only.
Use 14-16AWG wire for bus backbone. Terminal block or bus bar inside enclosure.

230V AC Live +5V DC +12V DC 18V AC GND / Neutral DNF = Do Not Fit . Dashed = Spare/Backup

PROJECT

G Scale Garden Railway Control
Power Supply Stack Detail

SHEET

D02 / D08
PS1, PS2, T1, Fuses, GND
Bus

REV

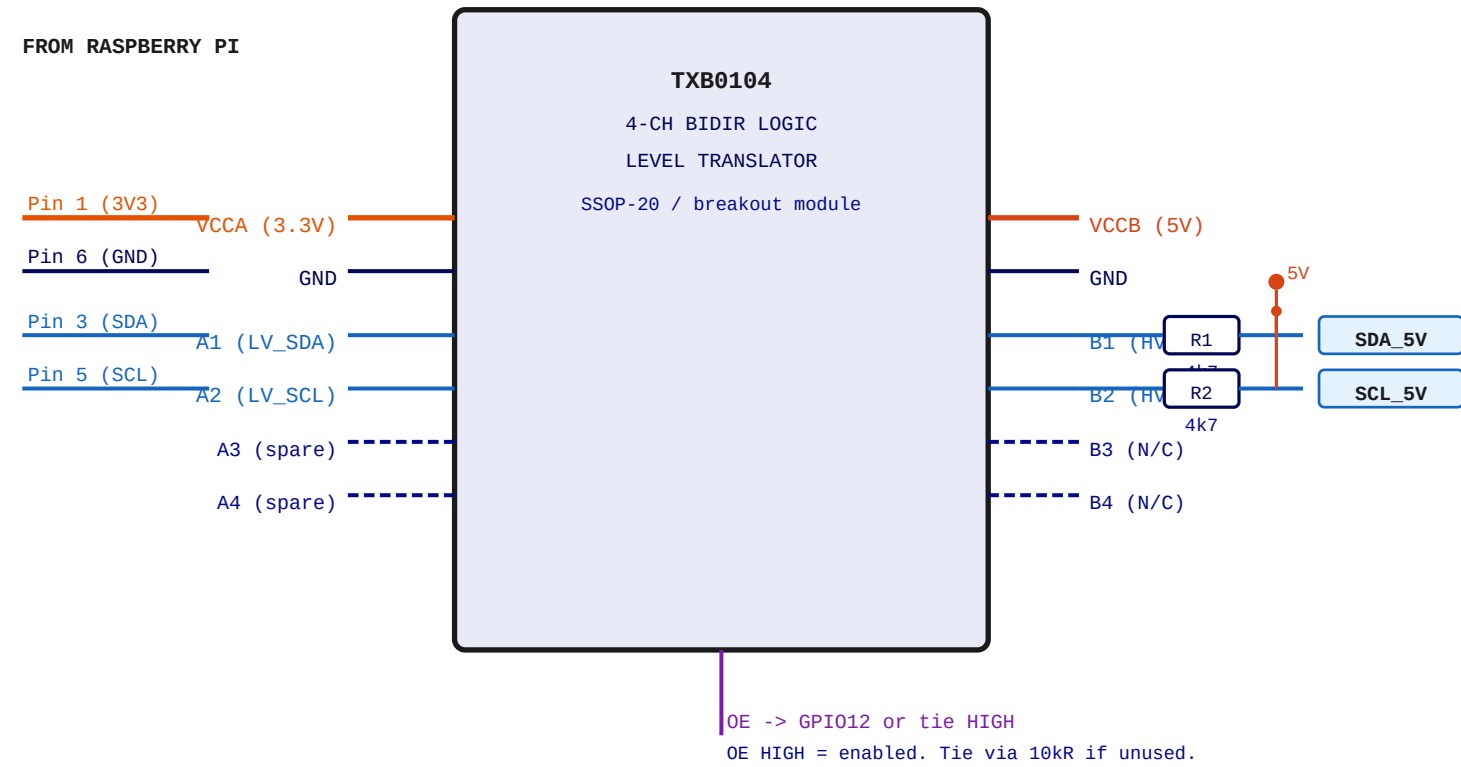
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DATE

2026-03

LEVEL SHIFTER - TXB0104 LS1 (ACTIVE) . LS2 (SPARE/DNF)

LS1 - TXB0104 (INSTALLED)



DECOUPLING

100nF between VCCA and GND
100nF between VCCB and GND
Place close to IC (C1 LV, C2 HV)

I2C BUS CAPACITANCE ANALYSIS

20m cable ~ 100pF/m x 2 cores = ~2000pF per wire
TXB0104 ~ 5pF, MCP23017 ~ 5pF x 4 = 20pF -> Total ~ 2020pF
Pull-up 4.7kR -> RC = 9.5us -> max ~50kHz at 20m
(!) For 400kHz: reduce to 1kR, verify with scope
(!) Pi 5 RP1 pull-ups on GPIO2/3 - may omit R1/R2 on short runs

LS2 - TXB0104 SPARE (DNF)



REPLACEMENT PROCEDURE

1. Power down. Verify 0V on GND bus.
 2. Note LS1 orientation (pin 1 marker).
 3. Remove LS1 from socket/header.
 4. Fit LS2 - same orientation.
 5. Power up. Verify i2cdetect: 0x20/21/22/40.
 6. Check SDA/SCL waveform with scope.
- Pack: x5 TXB0104 - 4 remain after LS2.

VCCA: 1.2-3.6V VCCB: 1.65-5.5V Channels: 4 Max: 100MHz
Bidirectional, no DIR pin. OE HIGH = outputs enabled.

(!) Pi 5: RP1 has internal pull-ups on GPIO2/3 (SDA/SCL)
Short runs (<2m): omit R1/R2. Long runs (20m): keep 4k7.

3.3V (VCCA) 5V (VCCB) SDA / SCL GND DNF = Do Not Fit . Pull-ups on HV (5V) side only

PROJECT
G Scale Garden Railway Control
Level Shifter TXB0104 Detail

SHEET
D03 / D08
LS1 Active, LS2 Spare

REV
C

DATE
2026-03

U4 - MCP23017 SPARE (0x23, DNF)

MCP23017

SPARE - DNF

Pre-addressed 0x23

A0 bridged, A1 bridged

A2 open

MCP23017 I/O EXPANDERS - U1 . U2 . U3 (ACTIVE) . U4 (SPARE/DNF)

ACTIVATION PROCEDURE

1. Address A0+A1 bridged -> confirmed 0x23

2. Connect VDD (5V), GND, SDA, SCL

3. Run i2cdetect - should show 0x23

4. Upgrade PS2 to 3A before activating LEDs

Adds channels: LED 41-56 or relay K7-K8 expansion

stored pre-addressed in labelled bag - verify with i2cdetect before soldering

FULL CHANNEL ALLOCATION - ALL MCP23017 UNITS

Unit	Port.Pin	Channel#	Function	Unit	Port.Pin	Channel#	Function	Unit	Port.Pin	Channel#	Function
U1	GPA.0	LED-01	Signal head 1 (Red)	U2	GPA.0-7	LED-17-24	Signal heads 7-10	U3	GPA.0-7	LED-33-40	Signal heads 15-18 / street lamps
U1	GPA.1	LED-02	Signal head 1 (Green)	U2	GPB.0-7	LED-25-32	Signal heads 11-14	U3	GPB.0	REL-K1	Turnout 1 (active LOW)
U1	GPA.2	LED-03	Signal head 1 (Amber)					U3	GPB.1	REL-K2	Turnout 2 (active LOW)
U1	GPA.3	LED-04	Signal head 2 (Red)					U3	GPB.2-5	REL-K3-K6	Turnouts 3-6 (active LOW)
U1	GPA.4-7	LED-05-08	Signal heads 2-3					U3	GPB.6-7	spare	Relay K7/K8 on spare module
U1	GPB.0-7	LED-09-16	Signal heads 4-6 + street lighting								
U4 (DNF)	GPA.0-7	LED-41-48	Expansion (future)								
U4 (DNF)	GPB.0-7	LED-49-56	Expansion or relay expansion								

DECOUPLING - 100nF MLCC cap between VDD and GND on each chip (4 caps total for U1-U3)

Place close to VDD pin. CJMCU-2317 breakout boards have pads for cap - fit 0805 or leaded 100nF.

(!) RELAY DEFAULT STATE - GPB on U3 must be initialised HIGH before any other output

At power-up all MCP23017 pins default LOW -> this energises relays. Fix: set all GPB HIGH in boot script before other init.

SPARES INVENTORY

U4 (0x23 pre-addressed) stored in labelled anti-static bag. Verify address with i2cdetect before soldering to board.

Additional MCP23017 can address 0x24-0x27 (A2 bridged combinations) for up to 8 devices = 128 I/O channels total on one I2C bus.

Stock: 1 spare CJMCU-2317 module in parts bag. AliExpress: "MCP23017 IIC I2C module".

GPA (LED outputs)

GPB (LED / relay)

GPB relay (active LOW)

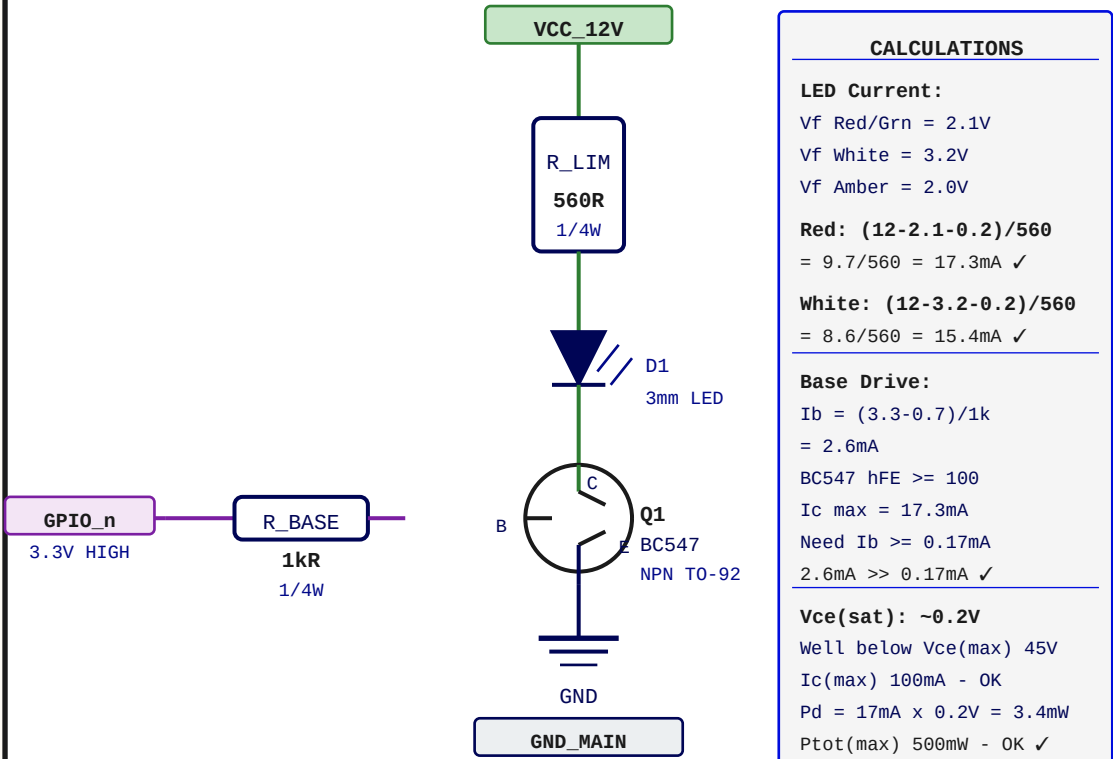
I2C Bus

DNF = Do Not Fit . ■ = address bit bridged . ■ = open

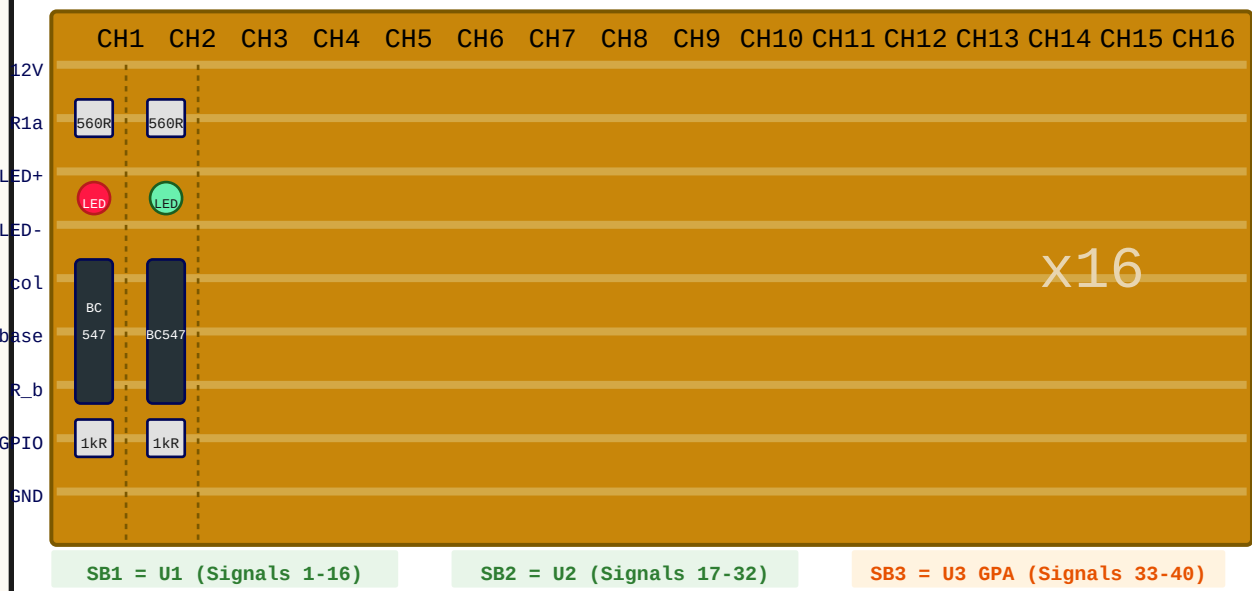
PROJECT	SHEET	REV	DATE	
G Scale Garden Railway Control	D04 / D08	C	2026-03	
MCP23017 I/O Expanders U1-U4	Expanders + Channel Alloc			

LED DRIVER STAGE - BC547 NPN - SINGLE CHANNEL DETAIL + STRIPBOARD

SINGLE CHANNEL - TYPICAL (x 48 CHANNELS TOTAL)



STRIPBOARD LAYOUT - SB1 (16 CHANNELS, TYPICAL x 3 BOARDS)



LED TYPE REFERENCE

Type	Vf	If	R_LIM (12V)	R_LIM (5V)
Red 3mm	2.1V	20mA	560R ✓	150R
Green 3mm	2.1V	20mA	560R ✓	150R
Amber 3mm	2.0V	20mA	560R ✓	150R
W.White 3mm	3.2V	15mA	560R ✓	120R

All values with 10% tolerance resistors confirmed in range.

SPARE COMPONENTS - BC547 DRIVER STACK

- x 50+ BC547 T0-92 (from 100-pack)
 - x 50+ 560R 1/4W (from 100-pack)
 - x 50+ 1kR 1/4W (from 100-pack)
 - x 1 Spare stripboard 9x15cm
 - x20+ each LED colour (high-brightness)
- All stored in labelled compartment box in enclosure.

VCC_12V / LED path

GPIO input

GND

Stripboard (copper side = brown)

PROJECT

G Scale Garden Railway Control
LED Driver Stack BC547 Detail

SHEET

D05 / D08
Single Channel +
Stripboard

REV

C

DATE

2026-03

BTS7960 43A H-BRIDGE MODULE



PWM PARAMETERS

Frequency: 5kHz recommended (1-20kHz range, tune per loco)

Forward: RPWM = 0-100% duty . LPWM = 0%

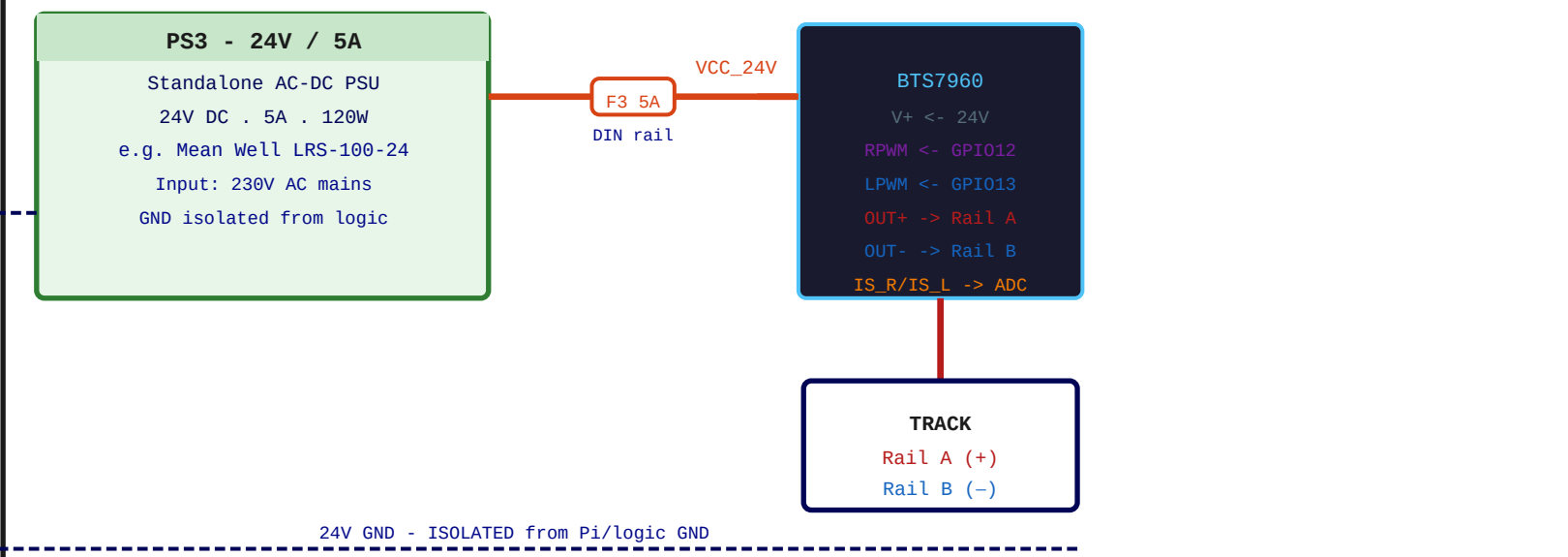
Reverse: RPWM = 0% . LPWM = 0-100% duty

Stop: both RPWM and LPWM = 0% (coasting) or both HIGH (brake)

```
# Pi 5 traction - pigpio (required on Pi 5)
import pigpio
pi = pigpio.pi()
def drive(pct): # -100=full rev, 0=stop, +100=full fwd
    dc = int(abs(pct) / 100 * 1000000)
    if pct >= 0: pi.hardware_PWM(12,5000,dc); pi.hardware_PWM(13,5000,0)
    else:       pi.hardware_PWM(12,5000,0); pi.hardware_PWM(13,5000,dc)
# drive(0)=stop drive(50)=fwd 50% drive(-30)=rev 30%
```

(!) Ramp speed gradually. Always drive(0) before reversing direction.
Add 500ms pause between forward and reverse commands in software.

TRACTION POWER + CURRENT SENSE



ADS1115 - 16-BIT I2C ADC - CURRENT SENSE

The diagram shows a central light blue box representing the ADS1115 chip. To the left, four pins are connected: VDD (3.3V) in orange, GND in dark blue, SCL -> Pi Pin5 in blue, and SDA -> Pi Pin3 in blue. To the right, four pins are connected: A0 <- IS_R (fwd current) in orange, A1 <- IS_L (rev current) in orange, A2 - spare in dashed blue, and A3 - spare in dashed blue. Inside the box, the following text is displayed: **ADS1115**, 16-bit . 4ch . I2C, Addr: 0x48, ADDR pin -> GND, PGA: +/-4.096V range, Rate: 128 SPS default, and Breakout board - I2C header.

ADS1115

16-bit . 4ch . I2C

Addr: 0x48

ADDR pin -> GND

PGA: +/-4.096V range

Rate: 128 SPS default

Breakout board - I2C header

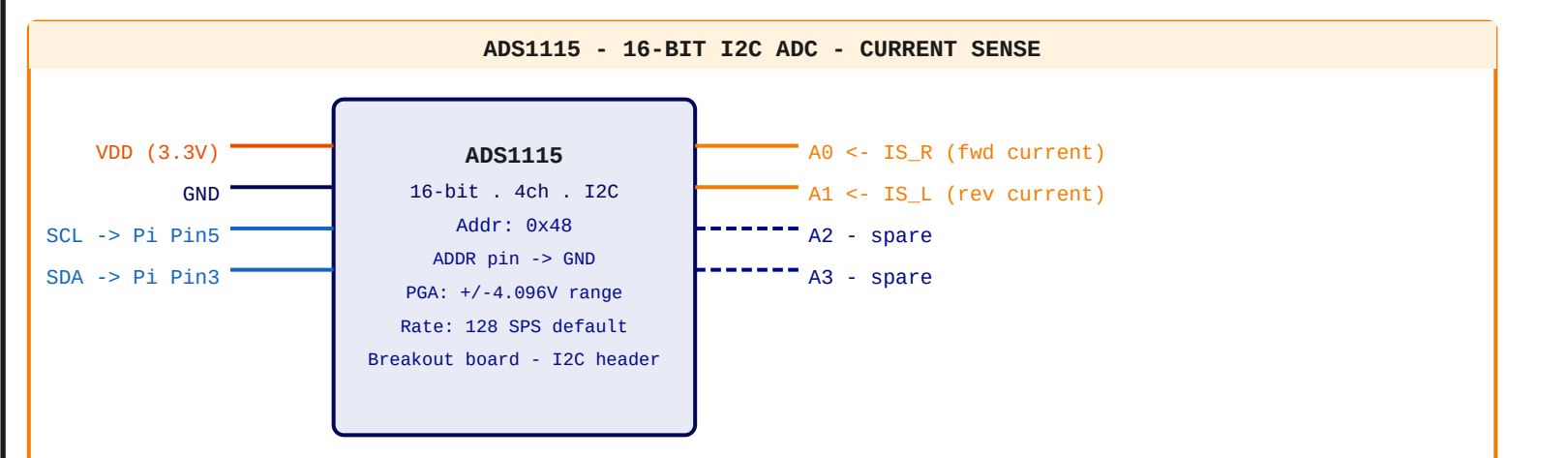
VDD (3.3V) — A0 <- IS_R (fwd current)

GND — A1 <- IS_L (rev current)

SCL -> Pi Pin5 — A2 - spare

SDA -> Pi Pin3 — A3 - spare

IS_R / IS_L output: current proportional to motor load. Add 1kR to GND on each IS pin before ADS1115 input.



IS_R / IS_L output: current proportional to motor load. Add 1kR to GND on each IS pin before ADS1115 input.

(!) 24V DC track GND is NOT shared with Pi/logic GND. ADS1115 IS inputs are signal-level only - no 24V on ADC pins.
BTS7960 IS pins output a small current (proportional to load) referenced to control GND - safe to connect directly to ADS1115 via 1kR.

I2C (SDA/SCL) VCC (5V logic) V+ (12V LED power) PWM output channels OE (output enable) DNF = Do Not Fit . Max 62 PCA9685 on one bus

PROJECT

G Scale Garden Railway Control

PCA9685 PWM Controller Detail

SHEET

D06 / D08

PWM1 Active, PWM2 Spare

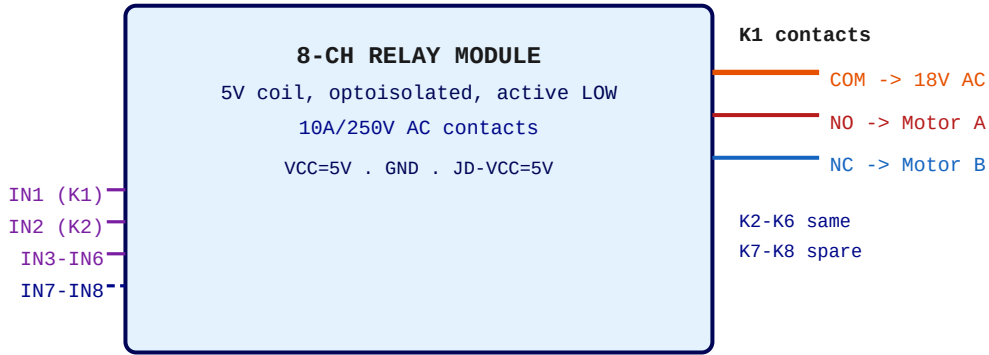
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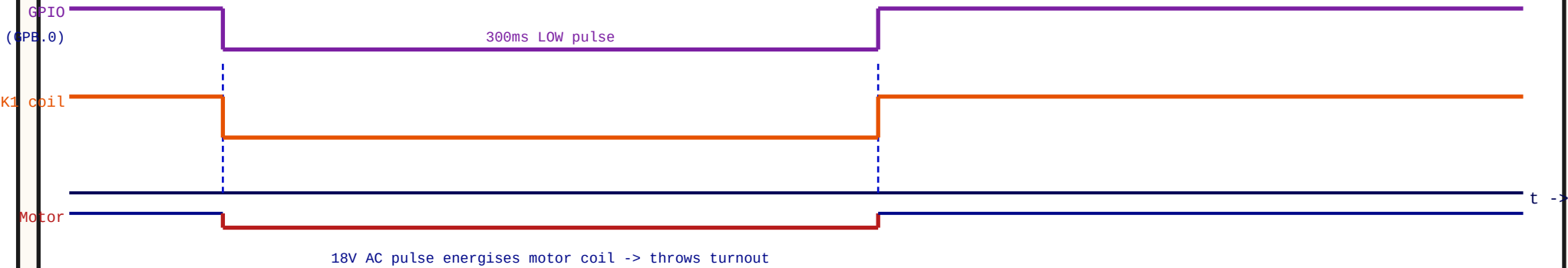
2026-03

RELAY / TURNOUT STAGE - 8CH MODULE . SPDT CONTACTS . INTERLOCK

8CH RELAY MODULE RM1 (ACTIVE)



RELAY TIMING DIAGRAM & INTERLOCK



(!) SOFTWARE INTERLOCK - MANDATORY

Never energise opposing relays for same turnout simultaneously.

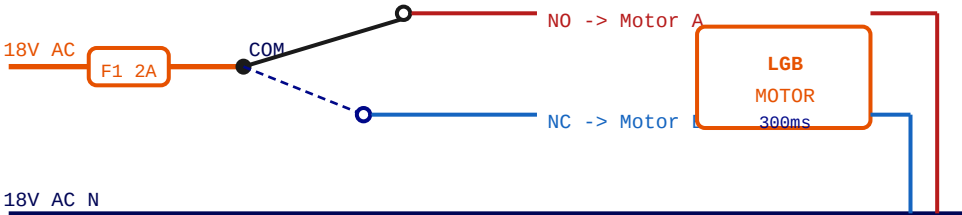
Although one relay per turnout (SPDT), if both K1 and K2 are wired to the same motor, energising both causes a short circuit.

Boot sequence requirement:

- Set ALL U3 GPB pins HIGH (deenergise) BEFORE any other init.
 - MCP23017 defaults LOW on power-up - this energises relays.
- Fix immediately in boot script before other initialisation.

```
# Boot script - run FIRST before anything else
for pin_num in range(8):
    mcp3.get_pin(8 + pin_num).value = True # GPB0-7 HIGH = relays OFF
# Pulse: set LOW 300ms then HIGH. Never hold LOW permanently.
# Interlock: never set K_odd and K_even for same turnout LOW together.
```

SPDT CONTACT (SINGLE TURNOUT)



SPARE MODULE RM2 (DNF)

Identical 8-ch 5V module. Pre-wired JD-VCC.
Swap time ~20min. Test: GPIO LOW -> hear click.
Label: "RM2 SPARE - DO NOT POWER BEFORE WIRING"
GPB6/GPB7 -> IN7/IN8 when K7/K8 activated.
Verify 18V AC is OFF during any relay swap.

(!) 18V AC ISOLATION - CRITICAL SAFETY NOTE

18V AC secondary is NOT connected to DC common ground bus.
Returns via transformer neutral only. F1 (2A) on AC live line.
Do not measure AC rail with DC GND reference - misleading readings.
Relay COM -> F1 -> 18V AC. NO/NC -> LGB motor terminals only.
Pi 5 GPIO active LOW (lgpio). Set pin HIGH = relay OFF (deenergised).

VAC_18V (COM) NO (motor A) NC (motor B) GPIO control (active LOW) 5V coil supply INIT ALL GPB HIGH ON BOOT

PROJECT
G Scale Garden Railway Control
Relay Turnout Stage Detail

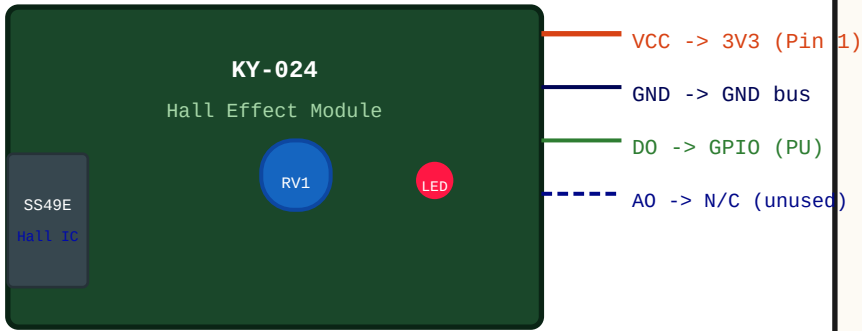
SHEET
D07 / D08
RM1 Active, RM2 Spare

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2026-03

HALL EFFECT SENSORS - KY-024 MODULE - PINOUT . BLOCK DETECTION

KY-024 MODULE - PINOUT & SCHEMATIC



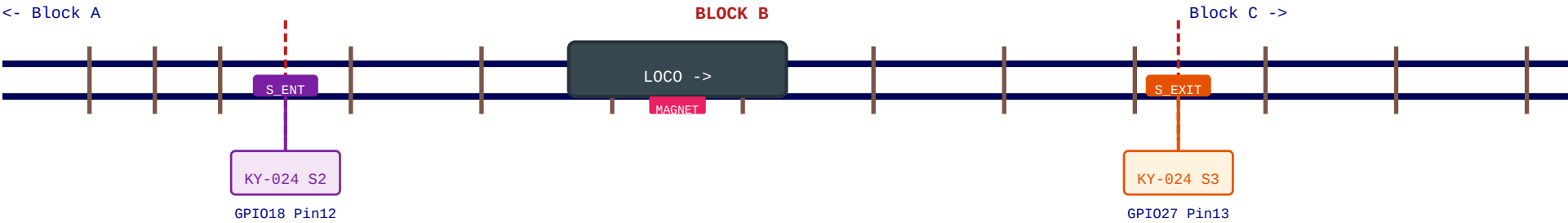
INTERNAL CIRCUIT (simplified)

3V3 -> SS49E VCC -> Hall IC -> comparator with RV1 -> digital out (DO)
A0 = analogue output proportional to field strength (not used here)
DO = LOW when magnet present, HIGH when absent (active LOW)
Pi 5 pull-up: set via lgpio.SET_PULL_UP - 50kR internal. No external resistor needed.
Current draw: ~8mA @ 3V3. Max cable run 10m at this voltage.

SENSITIVITY TRIMMER (RV1)

Turn CW to increase sensitivity. Adjust with magnet in position at operational distance. Confirm DO goes LOW when magnet aligned.
Seal trimmer with clear lacquer after setting (weather protection).

TWO-SENSOR BLOCK DETECTION - TRACK LAYOUT



BLOCK STATE MACHINE



Signal outputs per block state:

CLEAR -> Entry signal GREEN, exit signal GREEN
ENTERING -> Entry signal AMBER (caution)
OCCUPIED -> Entry signal RED, preceding block AMBER
EXITING -> Exit signal AMBER, entry signal AMBER
Signal logic implemented in software - this sheet is hardware only.
Minimum debounce: 50ms in software to ignore noise/vibration.
Magnet spec: 10x3mm neodymium N35 under loco chassis. Test trigger at layout height before finalising.

OUTDOOR INSTALL & WEATHERPROOFING

PCB Protection:

Apply clear acrylic lacquer or nail varnish to PCB body, trimmer pot, and solder joints.
DO NOT coat the SS49E sensor tip (black chip at board end) - will block magnetic field.
Seal wire entry points with silicone after lacquer cures.

Mounting:

Mount between sleepers, sensor end pointing upward. Magnet on loco should pass 2-5mm above sensor.
Use zip ties or small cable clips to board. Allow slight float - rigid mounting risks cracking in frost.
Sensor orientation: flat face of SS49E toward rail, perpendicular to motion.

Cable Runs:

3-core 22AWG stranded (VCC/GND/DO) max 10m at 3V3 with PUD_UP.
Use self-amalgamating tape at join points. Run in conduit if under lawn.
VCC: red 22AWG GND: black 22AWG DO: white/yellow 22AWG
Terminate at enclosure on screw terminal strip. Label each run.
If run exceeds 10m: add buffer (74HC14 Schmitt trigger at Pi end) or use 5V with level shifting.

SENSOR SPARES & TEST PROCEDURE

Spares Policy:

Purchase KY-024 in packs of 20. From a 20-pack:
8 installed (S0-S7), minimum 4 spare. Test ALL 20 before installing.
Label spares S_SPARE_01 etc. Store in foam-lined box, humidity sachet.
Failure modes: trimmer drift, PCB corrosion, SS49E mechanical damage.

Pre-installation test procedure:

1. Connect to Pi (3V3/GND/GPIO with PUD_UP)
2. Run (Pi 5 - lgpio): python3 -c "import lgpio; h=lgpio.gpiochip_open(0);
3. Pass magnet over sensor(h, 4, direction SET_PULL_UP(4) and not lgpio.gpio_read(h, 4))"
4. Adjust RV1 until trigger at ~5mm distance
5. Apply lacquer. Allow 1hr cure before mounting outdoors.
Magnet spec: 10x3mm N35 neodymium disc. Fix to loco chassis with epoxy.
One magnet per vehicle is sufficient. Multiple vehicles may use same magnet.

S_ENTRY (GPIO) S_EXIT (GPIO) 3V3 supply GND Magnet (N35) Do NOT seal SS49E sensor tip

PROJECT
G Scale Garden Railway Control
Hall Sensor KY-024 Detail

SHEET
D08 / D08
Sensors + Block Detection

REV
C

DATE
2026-03

Terminal Block Schedule (1 of 2) - POWER . TRACTION . TURNOUTS . SENSORS					
LEFT - TB1 POWER DISTRIBUTION . TB2 TRACTION			RIGHT - TB3 TURNOUT MOTORS . TB6 HALL SENSORS		
TB1 MAINS & POWER DISTRIBUTION					
FERRULE	LABEL	COL	DESCRIPTION	FROM / TO	
TB1:1	 BRN	L	230V AC Live	Mains -> F-MAINS -> PSU L	
TB1:2	 BLU	N	230V AC Neutral	Mains -> PSU N	
TB1:3	 GRN/YEL	E	Earth / PE	Mains -> enclosure bonding	
TB1:4	 RED	+5V	5V DC positive	PS1 -> Pi/logic/relay coils	
TB1:5	 BLK	0V	5V DC negative / logic GND	PS1 -> GND bus	
TB1:6	 RED	+12V	12V DC positive	PS2 -> LED stripboards	
TB1:7	 BLK	0V	12V DC negative	PS2 -> GND bus	
TB1:8	 RED	+24V	24V DC positive	PS3 -> F3 -> BTS7960 V+	
TB1:9	 BLK	0V-TRK	24V DC GND (isolated)	PS3 -> BTS7960 GND	
TB1:10	 ORG	18V~L	18V AC Live	T1 secondary -> F1 -> relay COM	
TB1:11	 ORG	18V~N	18V AC Neutral	T1 secondary -> motor returns	
TB2 TRACTION - TRACK OUTPUT (24V)					
FERRULE	LABEL	COL	DESCRIPTION	FROM / TO	
TB2:1	 RED	RAIL-A	Track Rail A (+)	BTS7960 OUT+ -> rail joiners	
TB2:2	 BLK	RAIL-B	Track Rail B (-)	BTS7960 OUT- -> rail joiners	
TB2:3	 RED	RAIL-A2	Track Rail A (2nd run)	Spare feed point if needed	
TB2:4	 BLK	RAIL-B2	Track Rail B (2nd run)	Spare feed point if needed	
TB3 TURNOUT MOTORS (18V AC)					
FERRULE	LABEL	COL	DESCRIPTION	FROM / TO	
TB3:1	 ORG	T1-A	Turnout 1 Motor term A	Relay K1 NO -> LGB motor	
TB3:2	 ORG	T1-B	Turnout 1 Motor term B	Relay K1 NC -> LGB motor	
TB3:3	 ORG	T2-A	Turnout 2 Motor term A	Relay K2 NO -> LGB motor	
TB3:4	 ORG	T2-B	Turnout 2 Motor term B	Relay K2 NC -> LGB motor	
TB3:5	 ORG	T3-A	Turnout 3 Motor term A	Relay K3 NO -> LGB motor	
TB3:6	 ORG	T3-B	Turnout 3 Motor term B	Relay K3 NC -> LGB motor	
TB3:7	 ORG	T4-A	Turnout 4 Motor term A	Relay K4 NO -> LGB motor	
TB3:8	 ORG	T4-B	Turnout 4 Motor term B	Relay K4 NC -> LGB motor	
TB3:9	 ORG	T5-A	Turnout 5 Motor term A	Relay K5 NO -> LGB motor	
TB3:10	 ORG	T5-B	Turnout 5 Motor term B	Relay K5 NC -> LGB motor	
TB3:11	 ORG	T6-A	Turnout 6 Motor term A	Relay K6 NO -> LGB motor	
TB3:12	 ORG	T6-B	Turnout 6 Motor term B	Relay K6 NC -> LGB motor	
TB6 HALL EFFECT SENSORS (3.3V)					
FERRULE	LABEL	COL	DESCRIPTION	FROM / TO	
TB6:1	 RED	3V3	Sensor supply 3.3V	Pi Pin 1 -> all KY-024 VCC	
TB6:2	 BLK	GND	Sensor GND	GND bus -> all KY-024 GND	
TB6:3	 VI0	S0-D0	Sensor 0 - Blk A ENT	KY-024 D0 -> GPIO4 Pin7	
TB6:4	 VI0	S1-D0	Sensor 1 - Blk A EXT	KY-024 D0 -> GPIO17 Pin11	
TB6:5	 VI0	S2-D0	Sensor 2 - Blk B ENT	KY-024 D0 -> GPIO18 Pin12	
TB6:6	 VI0	S3-D0	Sensor 3 - Blk B EXT	KY-024 D0 -> GPIO27 Pin13	
TB6:7	 VI0	S4-D0	Sensor 4 - Blk C ENT	KY-024 D0 -> GPIO22 Pin15	
TB6:8	 VI0	S5-D0	Sensor 5 - Blk C EXT	KY-024 D0 -> GPIO23 Pin16	
TB6:9	 VI0	S6-D0	Sensor 6 - Blk D ENT	KY-024 D0 -> GPIO24 Pin18	
TB6:10	 VI0	S7-D0	Sensor 7 - Blk D EXT	KY-024 D0 -> GPIO25 Pin22	
COLOUR KEY:  BRN=Live 230V  BLU=Neutral  GRN/YEL=Earth  RED=Positive DC  BLK=Negative/GND  ORG=18V AC/24V  GRN=Signal Grn  YEL=Signal Amb  WHT=St.Light  VI0=Sensor D0					
TB1 Power TB2 Traction TB3 Turnouts TB6 Sensors Sheet 1 of 2 - continued on D10					
PROJECT G Scale Garden Railway Control Terminal Block Schedule 1/2		SHEET D09 / D10 Power.Traction.Turnouts.Sensors		REV C	DATE 2026-03
NOTES TB1=Power TB2=Traction TB3=Turnouts TB6=Sensors. Continued D10.					

TERMINAL BLOCK SCHEDULE (2 of 2) - SIGNALS . STREET LIGHTS . BLOCK POWER

LEFT - TB4 LED SIGNAL OUTPUTS

RIGHT - TB5 STREET LIGHTING . TB7 BLOCK POWER (PENDING D06-B)

TB4 LED SIGNAL OUTPUTS (12V)						
FERRULE	LABEL	COL	DESCRIPTION		FROM / TO	
TB4:1	 RED	S1-R	Signal 1 Red		U1 GPA0 -> BC547 -> LED	
TB4:2	 GRN	S1-G	Signal 1 Green		U1 GPA1 -> BC547 -> LED	
TB4:3	 YEL	S1-A	Signal 1 Amber		U1 GPA2 -> BC547 -> LED	
TB4:4	 RED	S2-R	Signal 2 Red		U1 GPA3 -> BC547 -> LED	
TB4:5	 GRN	S2-G	Signal 2 Green		U1 GPA4 -> BC547 -> LED	
TB4:6	 YEL	S2-A	Signal 2 Amber		U1 GPA5 -> BC547 -> LED	
TB4:7	 RED	S3-R	Signal 3 Red		U1 GPA6 -> BC547 -> LED	
TB4:8	 GRN	S3-G	Signal 3 Green		U1 GPA7 -> BC547 -> LED	
TB4:9	 RED	S4-R	Signal 4 Red		U1 GPB0 -> BC547 -> LED	
TB4:10	 GRN	S4-G	Signal 4 Green		U1 GPB1 -> BC547 -> LED	
TB4:11	 YEL	S4-A	Signal 4 Amber		U1 GPB2 -> BC547 -> LED	
TB4:12	 RED	S5-R	Signal 5 Red		U1 GPB3 -> BC547 -> LED	
TB4:13	 GRN	S5-G	Signal 5 Green		U1 GPB4 -> BC547 -> LED	
TB4:14	 YEL	S5-A	Signal 5 Amber		U1 GPB5 -> BC547 -> LED	
TB4:15	 RED	S6-R	Signal 6 Red		U1 GPB6 -> BC547 -> LED	
TB4:16	 GRN	S6-G	Signal 6 Green		U1 GPB7 -> BC547 -> LED	
TB4:17	 BLK	COM-12	12V LED return		GND bus -> all LED cathodes	

TB5 STREET LIGHTING (12V)					
FERRULE	LABEL	COL	DESCRIPTION	FROM / TO	
TB5:1	 WHT	SL1	Street Light 1	U2 GPA4 -> BC547 -> warm white	
TB5:2	 WHT	SL2	Street Light 2	U2 GPA5 -> BC547 -> warm white	
TB5:3	 WHT	SL3	Street Light 3	U2 GPA6 -> BC547 -> warm white	
TB5:4	 WHT	SL4	Street Light 4	U2 GPA7 -> BC547 -> warm white	
TB5:5	 WHT	SL5	Street Light 5	U2 GPB4 -> BC547 -> warm white	
TB5:6	 WHT	SL6	Street Light 6	U2 GPB5 -> BC547 -> warm white	
TB5:7	 WHT	SL7	Street Light 7	U2 GPB6 -> BC547 -> warm white	
TB5:8	 WHT	SL8	Street Light 8	U2 GPB7 -> BC547 -> warm white	
TB5:9	 WHT	SL9	Street Light 9	U3 GPA6 -> BC547 -> warm white	
TB5:10	 WHT	SL10	Street Light 10	U3 GPA7 -> BC547 -> warm white	
TB5:11	 BLK	COM-SL	12V SL return	GND bus -> LED cathodes	

TB7 - BLOCK POWER SWITCHING (PENDING - Rev D)		
TB7:1		BLK-A+ - Block A Rail+ power
TB7:2		BLK-A- - Block A Rail- power
TB7:3		BLK-B+ - Block B Rail+ power
TB7:4		BLK-B- - Block B Rail- power
TB7:5		BLK-C+ - Block C Rail+ power
TB7:6		BLK-C- - Block C Rail- power
TB7:7		BLK-D+ - Block D Rail+ power
TB7:8		BLK-D- - Block D Rail- power

COLOUR KEY:  BRN=Live 230V  BLU=Neutral  GRN/YEL=Earth  RED=Positive DC  BLK=Negative/GND  ORG=18V AC/24V  GRN=Signal Grn  YEL=Signal Amb  WHT=St.Light VI0=Sensor DO

■ TB4 Signals
 ■ TB5 Street Lights
 ■ TB7 Block Power (pending)
 Sheet 2 of 2 - see D09 for TB1/TB2/TB3/TB6

PROJECT	SHEET	REV	DATE	NOTES
G Scale Garden Railway Control	D10 / D10	C	2026-03	
Terminal Block Schedule 2/2	Signals.Street Lights.Block Power			TB4=Signals TB5=Street Lights. TB7 Block Power pending Rev D / D06-B.